362

B.Sc (HONS.) IN CSE, FIRST YEAR, SECOND SEMESTER EXAMINATION, 2023

LINEAR ALGEBRA

[According to the New Syllabus]

Subject Code : 510225

Examination Code : 5612

Time—3 hours

Full marks—80

[N.B. The figures in the right margin indicate full marks. Answer any **four** *questions.]*

Marks

1. *(a)* Define degenerate and nondegenerate linear equations. Solve the following system of linear equation : 3+5 =8

$$x_1 - x_2 + 2x_3 = 5$$

$$2x_1 + x_2 - x_3 = 2$$

$$2x_1 - x_2 - x_3 = 4$$

$$x_1 + 3x_2 + 2x_3 = 1$$

(b) Determine the values of λ such that the following system of linear equations in x, y, z has *(i)* a unique solution *(ii)* no solution *(iii)* more than one solution : 6

$$x + y + \lambda z = 1$$

$$x + \lambda y + z = \lambda$$

$$\lambda x + y + z = \lambda^2$$

$$\text{CSD} = \sqrt{\frac{N_1\sigma_1^2 + N_2\sigma_2^2 + N_1 d_1^2 + N_2 d_2^2}{N_1+N_2}}$$

$$\bar{X} = \frac{N_1\bar{X}_1 + N_2\bar{X}_2}{N_1+N_2}$$

(c) Show that, 6

$$\begin{vmatrix} -1 & b & c & d \\ a & -1 & c & d \\ a & b & -1 & d \\ a & b & c & -1 \end{vmatrix} = (a+1)(b+1)(c+1)(d+1)\left[1 - \frac{a}{a+1} - \frac{b}{b+1} - \frac{c}{c+1} - \frac{d}{d+1}\right].$$

[Please turn over

4-19

upper

$A^T A$

$A^T A \cdot AA^{-1} = I$

Marks

2\. (a) Define: Upper and Lower triangular matrices, Orthogonal matrix, Idempotent matrix. — $2\times3=6$

(b) Prove that the matrix $A=\frac{1}{6}\begin{bmatrix}3 & 3 & 3 & 3\\ 3 & -5 & 1 & 1\\ 3 & 1 & 1 & -5\\ 3 & 1 & -5 & 1\end{bmatrix}$ is orthogonal. — 6

187

(c) Solve the following linear equations with the help of matrices : — 8

$3x + 5y - 7z = 13$
$4x + y - 12z = 6$
$2x + 9y - 3z = 20$

$X = A^{-1}B$

3\. (a) Define distance and norm in $\mathbb{R}^n$. Consider the vectors — 6

$\bar{x} = (1, -2, 3, -1)$, $\bar{y} = (2, 1, -5, 0)$ and $\bar{z} = (5, -3, -1, 1)$. Find

(i) $2\bar{x} + 3\bar{y} - 5\bar{z}$

(ii) $\bar{x}.\bar{y}$

(iii) $\|\bar{x} - \bar{y}\|$

(iv) $d(\bar{x}, \bar{z})$.

$SD = \cancel{\sqrt{\frac{1}{N}\sum f(x-\bar{x})^2}}$

$\sqrt{\frac{\sum(x-\bar{x})^2}{n-1}}$

Group SD $= \sqrt{\frac{\sum f(x-\bar{x})^2}{N}}$

(b) If $\bar{u}, \bar{v} \in \mathbb{R}^n$, then show that, — 8

$\|\bar{u}+\bar{v}\|^2 + \|\bar{u}-\bar{v}\|^2 = 2\left(\|u\|^2 + \|v\|^2\right)$ and give its geometric interpretation.

(c) If $\bar{u} = (1 + i, 2 - i)$ and $\bar{v} = (3 - 2i, 4 + i)$ then prove that, — 6

$\bar{u}.\bar{v} = \overline{\bar{v}.\bar{u}}$ and $d(\bar{u}, \bar{v}) = d(\bar{v}, \bar{u})$.

$MD = \frac{\sum|x-\bar{x}|}{n}$

4\. (a) Define norm of a vector. State and prove Cauchy-Schwarz theorem. — $2+6=8$

(b) Determine the value of α and β if x is orthogonal to both y and z, where $x = (1, \alpha, 2, -1)$, $y = (2, 1, 3, \beta)$ and $z = (\beta, 1, -4, 2)$. — 6

(c) For any vector $u, v, w \in C^n$ then prove that: — 6

(i) $(u + v). w = u.w + v.w$

(ii) $w. (u + v) = w. u + w.v$

with group MD $= \frac{\sum f|x-\bar{x}|}{N}$

$MD = \frac{\sum f|x-\bar{x}|}{N}$

5\. (a

6\.

$A^T A = I$

Marks

5. *(a)* Define linear transformation. Let $T : \mathbb{R}^3 \longrightarrow \mathbb{R}^3$ be given by $T(x, y, z) = (x, y, 0)\ \forall, x, y, z \in \mathbb{R}$. Show that T is a linear transformation. 2+5 =7

(b) Define rank and nullity of a linear transformation. If $T : u \longrightarrow v$ is a linear transformation then show that rank of T + nullity of $T = \dim u$. 2+6 =8

rank

(c) $T : \mathbb{R}^3 \longrightarrow \mathbb{R}^3$ be the linear operator defined by 5

$T(x, y, z) = (x + 2y - z, 2x - y + 4z, 4x + 3y - 2z)$.

Find a basis and the dimension of the *(i)* Im T of T, *(ii)* Ker T of T.

6. *(a)* Define : Inverse matrix, Echelon matrix, Spectrum. 2×3 =6

(b) By using row canonical form find A^{-1} and hence or otherwise solve the following equation: 6

$$A\begin{pmatrix} w \\ x \\ y \\ z \end{pmatrix} = \begin{pmatrix} 0 \\ 7 \\ 4 \\ 6 \end{pmatrix} \text{ when } A = \begin{pmatrix} 0 & -1 & -2 & -3 \\ 1 & 1 & 4 & 4 \\ 1 & 3 & 7 & 9 \\ 1 & -2 & -4 & -6 \end{pmatrix}$$

(c) Find the all Eigen values and the corresponding eigen vectors for the matrix : $A = \begin{pmatrix} 3 & 1 & 1 \\ 2 & 4 & 2 \\ 1 & 1 & 3 \end{pmatrix}$. 8

Is the matrix diagonalizable? Also find an invertible matrix P such that $P^{-1}AP$ is a diagonal matrix.

$\frac{\Sigma(x-\bar{x})^2}{n}$

Sample variance = $\frac{\Sigma(x-\bar{x})}{n-1}$

$A^{-1} = \frac{1}{|A|} \text{adj } A^T$

B.Sc. (HONS.) IN CSE FIRST YEAR, SECOND SEMESTER EXAMINATION, 2021

DISCRETE MATHEMATICS

[According to the New Syllabus]

Subject Code : 510223

Examination Code : 5612

Time—3 hours

Full marks—80

[N.B. The figures in the right margin indicate full marks. Answer any **four** *of the following questions.]*

Marks

1. (*a*) Which of the following sentences are proposition? What are truth values of those that are propositions? 4
   (*i*) Savar is the capital of Bangladesh.
   (*ii*) Is he ill?
   (*iii*) $x + 2 = 5$.
   (*iv*) You pass the course.

   (*b*) Let $P(x)$, $Q(x)$ and $R(x)$ be the statements "x is a lion", "x is fierce" and "x drinks coffee" respectively. Assuming that the universe of discourse is the set of all creatures. Express each of the following statements using quatifiers, logical connectives and $P(x)$, $Q(x)$ and $R(x)$: 8
   (*i*) All lions are fierce.
   (*ii*) Some lions do not drink coffee.
   (*ii*) Some fierce creatures do not drink coffee.

   (*c*) Consider the following propositions : 4
   p : You can send an E-mail.
   q : Your inbox is full. $r \wedge \neg q$.
   r : New e-mails are stored in your inbox.
   Translate the following sentences into logical expressions :
   (*i*) You cannot send an e-mail if and only if your inbox in full.
   (*ii*) New e-mails are stored in your inbox unless your inbox is full.

   (*d*) Show that $(p \wedge q) \longrightarrow (p \vee q)$ is a tautology. 4

2. (*a*) Define set and power set. What is the power set of $\{a, b, c\}$? 5

   (*b*) Let $A_i = \{1, 2, 3, \ldots\ldots\ldots\ldots, i\}$. Then $\bigcup\limits_{i=1}^{n} A_i =$ and? $\bigcap\limits_{i=1}^{n} A_i = ?$ 5

   (*c*) Define composition of function. Let f and g be the function from the set of integers to the set of integers defined by the $f(x) = 4x + 2$ and $g(x) = 2x + 1$. What is the composition of f and g? What is the composition of g and f? 5

[Please turn over

Marks

(*d*) What is function? Is the following figure a function? Why is it or not? 5

3. (*a*) What is relation? Write down the properties of relation. 5

(*b*) Let $R = \{(1, 1), (2, 1), (3, 2), (4, 3)\}$. Find the powers $R^n, n = 2, 3, 4, \ldots\ldots\ldots\ldots$ 5

(*c*) Consider the following relation on $\{1, 2, 3, 4\}$: 5
$R = \{(1, 1), (1, 2), (1, 4), (2, 1), (2, 2), (3, 3), (4, 1)\}$. Is the relation reflexive, symmetric and transitive?

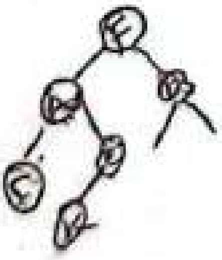
(*d*) Find the matrix representation of the relations SoR where the matrices representing R and S are $M_R = \begin{bmatrix} 1 & 0 & 1 \\ 1 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix}$ and 5

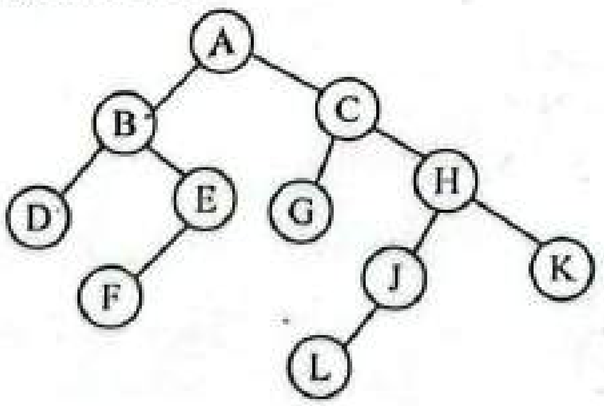
$M_S = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & 0 & 1 \end{bmatrix}$.

4. (*a*) Draw the Hasse diagram representing the partial ordering $\{(a, b) \mid a \text{ divides } b\}$ on $\{1, 2, 3, 4, 6, 8, 12\}$. 5

(*b*) Draw a binary tree from the traversal : 5
Inorder : E A C K F H D B G
Preorder : F A E K C D H G B

(*c*) Write down the preorder, inorder and postorder traversing for the following binary tree : 5

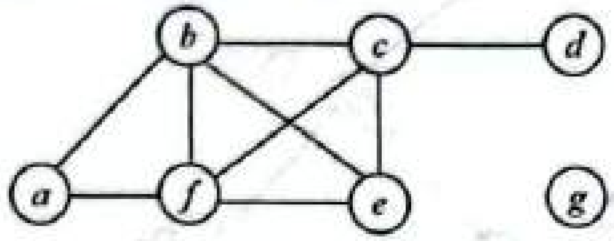




(*d*) Evaluate the prefix expression $+ - * 2 3 5 / \uparrow 2 3 4$? 5

Marks

5. (*a*) What is graph? What is the degree of vertices in the graph G displayed in following figure : 5



(*b*) Draw the precedence graph for the following expression : 5

$S1$: $x := 0$
$S2$: $x := x + 1$
$S3$: $y := 2$
$S4$: $z := y$
$S5$: $x := x + 2$
$S6$: $y := x + z$
$S7$: $z := 4$

(*c*) Define chromatic number. What is the chromatic number of $K_{3,4}$ and K_5? 5

(*d*) Draw a graph with the adjacency matrix : 5

$$\begin{bmatrix} 0 & 1 & 1 & 0 \\ 1 & 0 & 0 & 1 \\ 1 & 0 & 0 & 1 \\ 0 & 1 & 1 & 0 \end{bmatrix}$$

6. (*a*) What is mathematical induction? Using mathematical induction prove that the sum of the first n odd positive integers is n^2. 5

(*b*) Define the Sum Rule and the Product Rule. How many different bit strings are there of length eight? 5

(*c*) Describe the universality of NAND and NOR gate. 5

(*d*) Simply the Boolean function $F = \overline{ABC} + \overline{B}C\overline{D} + \overline{A}BC\overline{D} + \overline{ABC}$ using K-map. 5

BSc (HONS.) IN COMPUTER SCIENCE AND ENGINEERING

FIRST YEAR, SECOND SEMESTER EXAMINATION, 2020

[According to the New Syllabus]

LINEAR ALGEBRA

Subject Code : CSE-510225

Examination Code : 5612

Time-3 hours

Full marks--80

(N.B. The figures in the right margin indicate full marks. Answer any four questions.]

1. (a) Define homogeneous and non-homogeneous linear equation. 3+5=8

Reduce the following system of linear equations into echelon form and solve it.

$$\begin{aligned} x_1 - x_2 + 2x_3 &= 5 \\ 2x_1 + x_2 - x_3 &= 2 \\ 2x_1 - x_2 - x_3 &= 4 \\ x_1 + 3x_2 + 2x_3 &= 1 \end{aligned}$$

(b) Determine the values of λ such that the following system of linear equations in unknowns x, y and z has (i) a unique solution (ii) no solution (iii) more than one solution :

$$\begin{aligned} x + y + \lambda z &= 1 \\ x + \lambda y + z &= \lambda \\ \lambda x + y + z &= \lambda^2 \end{aligned}$$

(c) If $2s = a + b + c$ then prove that,

$$\begin{vmatrix} a^2 & (s-a)^2 & (s-a)^2 \\ (s-b)^2 & b^2 & (s-b)^2 \\ (s-c)^2 & (s-c)^2 & c^2 \end{vmatrix}$$

2. (a) Define : Singular Matrix, Nilpotent Matrix and Hermitical Matrix 2x3=6.

(b) If A and B are non-singular matrices, then prove that, 7

$$(AB)^{-1} = B^2A^2$$

(c) Define the rank of a matrix. Find the rank of the following matrix : 2+5=7

$$\begin{bmatrix} 1 & 2 & 0 & -1 \\ 2 & 6 & -3 & -3 \\ 3 & 10 & -6 & -5 \end{bmatrix}$$

3. (a) If A = $\begin{bmatrix} 3 & -3 & 4 \\ 2 & -3 & 4 \\ 0 & -1 & 1 \end{bmatrix}$ prove that, $A^2 = A^{-1}$

(b) Show that the matrix A= $\begin{bmatrix} \cos\theta & \sin\theta & 0 \\ -\sin\theta & \cos\theta & 0 \\ 0 & 0 & 1 \end{bmatrix}$

is invertible for all values θ and find A^{-1}.

4. (a) Define linear transformation let T: 13 ~>R3 be given by 2+5=7
$T(a,b,c)$ _(a,b,o) $\forall a,b,c \in R$.. Show that T is a linear transformation.
(b) Define rank and nullity of a linear • transformation. If $T: U \to V$ is a linear transformation then show that rank of $T +$ *nullity of* $T \doteq \dim U$ 2+6=8
(c) $T: R^3 \to R^3$ be the linear operator defined by
$T(x,y,z) = (x + 2y - z, 2x - y + 4z, 4x + 3y - 2z)$.
Find a basis and the dimension of the (i) Im T of T
(ii) K.er T of T.

5. (a) Define matrix representation of a linear operator. Let $\{u_1,, u_2, ... U_m\}$ be a basis of Uand T be any linear operator of U, then for any vector $W \in U$, 10
Show that $[T]_u\ [W]_u = [T\ (W)]_u$.
(b) The mapping $T: R^3 \to R^3$ is defined by 10
$T\,(x,y,z)$ _$(2y + z, x - 4)), 3x)$.
Show that, T is a linear transformation and find the matrix T in the basis $\{u_1 =$ _$(1,1,1), u_2 = (1,1,0), u_3 = (1,0,0)\}$.

6. (a) Define Eigen values and Eigen vectors. State and prove the Clayey -Hamilton theorem. 2+6=8
(b) Find the Eigen values and the corresponding Eigen vectors of 12
the matrix $A = \begin{bmatrix} 3 & 1 & 1 \\ 2 & 4 & 2 \\ 1 & 1 & 3 \end{bmatrix}$
Also verify Cayley-Hamilton theorem for the matrix A.

B.Sc (HONS.) IN COMPUTER SCIENCE AND ENGINEERING FIRST YEAR SECOND SEMESTER EXAM -ATION, 2019

[According to the New Syllabus

CSE-510225

(Linear Algebra)

Time-3 hours

Full marks-80

[N.B. The, figures fn the right margin indicate full marks. Answer any four questions.]

Marks

1. (a) Define linear equation with examples. 2
(b) Determine the value of k such that the following system of linear equations in unknowns x, y, z has : 6
(i) a unique solution (ii) no solution (iii) more than one solutions.
(c) Define norm of a vector. Consider the points, P (3, k, -2) and Q(5, 3, 4) in R^3 . Find the value of λ so that $\overrightarrow{PQ}$ is orthogonal to the vector (4, -3, 2). 6

(d) Prove that, $\begin{bmatrix} -a & -b & c & d \\ b & -a & -d & c \\ c & -d & a & -b \\ d & c & b & a \end{bmatrix} = (a^2 + b^2 + c^2 + a^2)^2.$ 6

2. (a) Define (i) symmetric matrix (ii) square matrix (iii) idempotent matrix (iv) singular matrix. 4

(b) If A and B are comparable matrices and AT and BT are the transpose matrices of A and B respectively, then prove that 1+1+4 =6
(i) (AT)T =A, (ii) (A + B)T =AT ± BT, (iii) (AB)T = BTAT.

(c) Solve the following system of linear equations with the help of matrix : 5
x+2y+3z+4=0
2x+4y+5z+7=0
3x+5y+6z+ 10=0

(d) Show that the matrix $A=\begin{bmatrix} 2 & -1 & 1 \\ -2 & 3 & -2 \\ -4 & 4 & -3 \end{bmatrix}$ is idempoterrt. 5

3. (a) Define linear transformation. Show that the product of two linear transformation is a linear transformation. 2+5=7

(b) Let S and T be the linear operators of RZ into RZ defined by S(x,Y)=3x+2y,-6x+y) T(x,Y)=(2x+y,x-y). Find formulae defining properties S+ 7; ST, TS and S2. 7

(c) Define (i) System of linear equation 6
(ii) Consistent
(iii) Homogeneous
(iv) Non-homogeneous

4. (a) Define minors and co-factors with example. 3

(b) If $\Delta = \begin{vmatrix} x & x^2 & x^3+1 \\ y & y^2 & y^3+1 \\ z & z^2 & z^3+1 \end{vmatrix}$ I then prove that, 6

A=(x -y) (y -z) (z -x) (xyz - l). Also show that if x, y and z are not equal and A = 0, then xyz + 1- 0.

(c) Prove that,

$$\begin{bmatrix} 1 & a & a^2 & a^3 \\ 1 & b & b^2 & b^3 \\ 1 & c & c^2 & c^3 \\ 1 & d & d^2 & d^3 \end{bmatrix}$$

5. (a) Define image and kernel of a linear transformation. 4

(b) Show that, T: R3 -> R2, where T (x, y, z) _(x + y - z, 2x - y+ 2z) is a linear transformation. Find a basis and dimension for ImT =l 1 and ker T. 1 1

(c) Find a linear transformation T : lt3 _> 1};4 whose ImT is generated by I(l, 2,0,-4),(2, 0,-], -3)}. 5

6. (a) Define characteristic matrix and characteristic equation. 3

(b) Find the eigen values and eigen vectors of the matrix 8

$$A=\begin{bmatrix}1 & 2 & 2\\ 1 & 2 & -1\\ -1 & 1 & 4\end{bmatrix} \quad 4$$

(c) State Cayley-Hamilton theorem. Using this theorem find the 2+7

inverse of the matrix $A = = \begin{bmatrix}2 & 0 & -1\\ 5 & 1 & 0\\ 0 & 1 & 3\end{bmatrix} \quad =9$

B. Sc (HONS.) IN CSE PART-I, SECOND SEMESTER EXAMINAT'ION, 2018

[According to the New 3~yllahusJ

CSE-510211

Examination Code : 5612

(Linear Algebra)

Time-3 hours

Full marks-80

1. (a) Define : Singular 'matrix, Nilpotent matrix and Hermitian matrix. 2x3=6

(b) If A and B are non-singular matrices, then prove that, (AB) $= B^{-1}A^{-1}$ Also prove that, $(A^{-1})^{-1} = A$ and $(A^{-1})^{t} = (A^{t})^{-1}$. 7

(c) Define rank of matrix. Find the rank of the following matrix : 2+5=7

$$A = \begin{matrix}1\,2 & 0 & -1\\ 2\,6 & -3 & -3\\ 3\,10 & -6 & -5\end{matrix}$$

2. (a) Define, inverse matrix. If $A = \begin{matrix}1 & -1 & 1\\ 2 & -1 & 0\\ 1 & 0 & 0\end{matrix}$ then find $A^{-1} + 2A^{t}$. $(2 + 8 = 10)$

(b) Show then $\begin{vmatrix}-1 & b & c & d\\ a & -1 & c & d\\ a & b & -1 & d\\ a & b & c & -1\end{vmatrix}$

$= (a+1)(b+1))c+1)(d+1)\left[1 - \frac{a}{a+1} - \frac{b}{b+1} - \frac{c}{c+1}, - \frac{d}{d+1}\right]$

3. (a) Solve the following system of linear equations with the help of matrix: 8

$x + 2y + z = 2$

$2x - y + 2z = -1$

$3x - 4y - 3z = -16$

(b) (i) Solve the system of linear equations : 6

$2x + y - 2z = 10$

$3x + 2y + 2z = 1$

$5x + 4y + 3z = 4$

(ii) Find the non-trivial solution for the system of linear equations: 6

$x + 2y - 3z = 0$

$2x + 5y = 2z = 0$

$3x - y - 4z = 0$

4. (a) Define degenerate and non-degenerate linear equation. 2

(b) Determine the values of λ such that the following system of 6 linear equations in unknowns x, y and z has (i) a unique solution (ii) no solution (iii) more than one solution:

$$x + y + \lambda z = 1$$
$$x + \lambda y + z = \lambda$$
$$\lambda x + y + z = \lambda^2$$

(c) Show that, $\begin{vmatrix} a+b+c & a+b & a & a \\ a+b & a+b+c & a & a \\ a & a & a+b+c & a+b \\ a & a & a+b & a+b+c \end{vmatrix}$ 6

$= c^2(4a+2b+c)(2b+c)$.

(d) Define norm of a vector. Consider P(3, λ-2) and Q(5, 3, 4) 6
in $\mathbb{R}^3$. Find the value of λ so that, $\overrightarrow{PQ}$ is orthogonal to the vector (4, -3, 2).

5. (a) Define linear transformation. Let U and V be two vector spaces over 2+5=7
the field F and T_1 and T_2 be linear transformations from U into V then prove that, $T_1 + T_2$ is a linear transformation.

(b) If S and T be the linear transformation $S, T: \mathbb{R} \longrightarrow \mathbb{R}^3$ defined by 8
S(x, y, z) =(y, z, x) and T(x, y, z) =(z + y+ z, 0, 0) then find (i) (TS) (1, 0, 1), (ii) (ST) (1, 0, 1) (iii) (S + T) (1, 0, 1). (iv) (S- T) (1, 0, 1).

(c) Show that the mapping $T: \mathbb{R}^2 \longrightarrow \mathbb{R}^3$ defined by $T(x_1, x_2) = (x_1 + x_2, x_1 - x_2, x_1)$ is linear.

6. (a) Define eigen values and eigen vectors. State and prove 2+6=8
Cayley-Hamilton theorem.

(b) Find the eigen values and the corresponding eigen vectors of 12
the matrix $A = \begin{bmatrix} 3 & 1 & 1 \\ 2 & 4 & 2 \\ 1 & 1 & 3 \end{bmatrix}$

Also verify the Cayley-Hamilton theorem for the matrix A.